Academic year 2025-2026 (Odd Sem)

CIE-1 - SCHEME AND SOLUTION

Course Title	**Database Management Systems**	
Course Code	CD252IA	
Sem	V	

Sl.No	PART-A	M
1.	In the ENROLLMENT table, which represents a **weak entity**, the **partial key** is the combination of **CourseID** and **Semester**.	**2**
2.	• A database can provide different views (subsets or formats) of the same data to different users based on their requirements. • Each view presents only relevant data and hides unnecessary or sensitive information. In a **Library Management System**: • A **librarian** views details of all books, members, fines, and issue records. • A **student** can view only their own borrowed books and due dates.	**2**
3.	Yes, the **Prescription** entity can be considered a **weak entity** because: • It **depends on both the Patient and Doctor (or Consultation)** entities for its existence. • The **Prescription** does not have a unique primary key of its own; instead, it uses a **composite key** formed by **Patient_ID** and **Consultation_ID** to identify each record uniquely. • If the associated **Consultation** record is deleted, the corresponding **Prescription** records lose their meaning and must also be deleted.	**2**
4.	• **Multivalued Attribute:** An attribute that can have multiple values for a single entity. *Example:* Phone numbers of an employee — {Phone: 9876543210, 9123456789} • **Composite Attribute:** An attribute that can be divided into smaller subparts with independent meaning. *Example:* Address → (House_No, Street, City, Pincode)	**2**
5.	NULL represents missing, unknown, or inapplicable information in a database. It allows differentiation between "no value provided" and "value = 0 or blank," thereby maintaining data integrity and flexibility in record keeping. **Instances:** 1. **Employee table:** Attribute *Date_of_Resignation* can be NULL for currently employed staff. 2. **Student table:** Attribute *Scholarship_Amount* can be NULL if the student has not been granted any scholarship.	**2**

Sl. No.		PART-B			M
1		**Operation**	**Possible Constraint Violations**	**Enforcement Mechanisms**	**10**
		1. Deleting a Project	Deletion violates referential integrity if tuples in `WORKS_ON` reference the deleted project.	Use `ON DELETE CASCADE` or `ON DELETE SET NULL` in foreign key (WORKS_ON.Pno → PROJECT.Pnumber).	
		2. Updating Manager Details	Updating `Mgr_ssn` may break referential integrity if the new SSN does not exist in `EMPLOYEE`.	Foreign key constraint (`Mgr_ssn` → `EMPLOYEE.Ssn`) with referential integrity enforcement.	
		3. Inserting Employee	`Dno` must refer to an existing department; `Ssn` must be unique.	Primary key on `Ssn`, foreign key (`Dno` → `DEPARTMENT.Dnumber`).	
		4. Assigning Manager to Department	Manager must be an employee in that department; avoid circular references.	Use check constraint ensuring `Mgr_ssn` ∈ `EMPLOYEE.Ssn` and triggers for department association.	
		5. Updating Work Hours	Hours cannot be negative or exceed allowed limit; employee and project IDs must exist.	Check constraint on `Hours >= 0`, foreign keys (`Essn` → `EMPLOYEE.Ssn`, `Pno` → `PROJECT.Pnumber`).	

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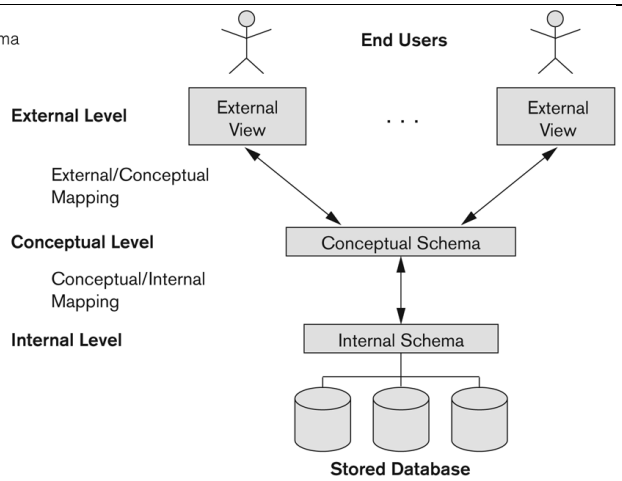
2	**(a)**	• The operation violates the **Referential Integrity Constraint (Foreign Key Constraint)** because the new Dept_ID ('D999') does not exist in the DEPARTMENT table. – Explain referential integrity constraint • The DBMS enforces this constraint to ensure that every Dept_ID in the STUDENT table corresponds to a valid department.	**06**
	(b)	Referential integrity ensures that a foreign key in one table always refers to an existing primary key in another table. When a delete operation is performed on a record that is referenced elsewhere, it can cause a violation of this integrity rule. Other types of constraints (like domain, key, or entity integrity) are not affected directly by a delete operation. DEPARTMENT(Dno, Dname) EMPLOYEE(Eid, Ename, Dno) Here, EMPLOYEE.Dno is a foreign key referencing DEPARTMENT.Dno. If a department with Dno = 10 is deleted from the DEPARTMENT table while employees still exist with Dno = 10 in EMPLOYEE, the Referential Integrity constraint is violated because those employees now reference a non-existent department.	**04**
3	**(a)**	**Figure 2.2** The three-schema architecture.  Diagram – 03 marks, Explanation-03 marks	**06**

	(b)	• **Logical Data Independence:**
		  • The capacity to change the conceptual schema without having to change the external schemas and their associated application programs.
		• **Physical Data Independence:**
		  • The capacity to change the internal schema without having to change the conceptual schema.
		  • For example, the internal schema may be changed when certain file structures are reorganized or new indexes are created to improve database performance
			04

4		Identifying entities, attributes, relationships, types of attributes – 06 marks
		Depiction using ER notations – 04 marks
		Possible ER components (but not limited to)
		Entities and Attributes:
		1. **STUDENT**
		  o *Student_ID* (Primary Key)
		  o Name
		  o Email
		  o Department_ID *(Foreign Key → DEPARTMENT.Department_ID)*
		2. **FACULTY**
		  o *Faculty_ID* (Primary Key)
			10

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- Name
- Email
- Department_ID *(Foreign Key → DEPARTMENT.Department_ID)*

3. **DEPARTMENT**
   - *Department_ID* (Primary Key)
   - Dept_Name
   - Head_of_Department
4. **COURSE**
   - *Course_ID* (Primary Key)
   - Course_Name
   - Credits
   - Department_ID *(Foreign Key → DEPARTMENT.Department_ID)*
5. **ENROLLMENT**
   - *Enrollment_ID* (Primary Key)
   - Student_ID *(Foreign Key → STUDENT.Student_ID)*
   - Course_ID *(Foreign Key → COURSE.Course_ID)*
   - Enrollment_Date
   - Grade
6. **HANDLES**
   - *Faculty_ID (Foreign Key → FACULTY.Faculty_ID)*
   - *Course_ID (Foreign Key → COURSE.Course_ID)*

Possible
Multivalued attribute: Phone_Number
Composite attribute: Address

Relationships:

Relationship	Type	Description
Department – Student	1 : M	A department has many students
Department – Faculty	1 : M	A department has many faculty members
Department – Course	1 : M	A department offers many courses
Student – Enrollment – Course	M : N	Students can enroll in many courses and eacl course can have many students
Faculty – Handles – Course	M : N	Faculty can handle multiple courses and eacl course can be handled by multiple faculty members

Assumptions

1. A **student** belongs to **only one department**.
2. Each **faculty** belongs to **one department**.
3. A **course** is offered by **only one department**.
4. The **ENROLLMENT** table resolves the M:N relationship between *Student* and *Course*.
5. The **HANDLES** table resolves the M:N relationship between *Faculty* and *Course*.
6. **Grade** can be NULL until the course is completed.

5		Applying Algorithm to the above ER Diagram – 07 marks Using appropriate notations to depict schema – 03 marks	**10**

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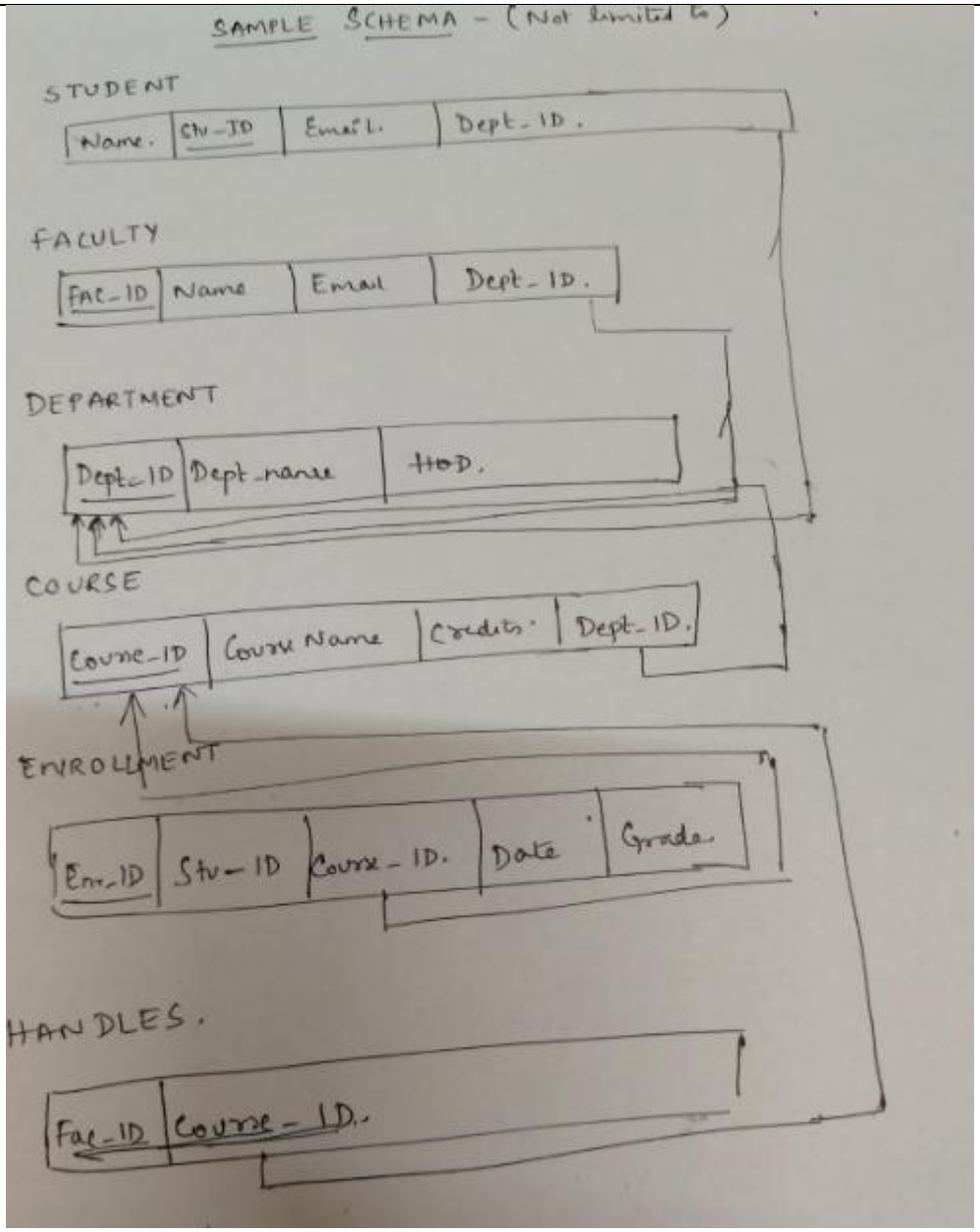
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